MRI Super Resolution Using ESRGAN

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***Abstract*— Magnetic resonance imaging (MRI) at high spatial resolution is essential for accurate detailed and quantitative physiologic assessment. However, single image super resolution (SISR) techniques with traditional long scan times, reduced spatial masking, and low signal-to-noise ratio (SNR) to acquire MR images with high spatial resolution have emerged as a solution promising high-resolution reproduction from relatively simple MR images. In this work, we introduce a new method called ESRGAN. ESRGAN builds on the GAN framework and improves its design and operation. Key features of ESRGAN include optimized network design, improved adversary loss functions, and sensitivity loss integration for image quality.**

Keywords— Magnetic Resonance Images (MRI), Signal-to-Noise Ratio (SNR), Single Image Super-Resolution (SISR), Enhanced Super-Resolution Generative Adversarial Network (ESRGAN)

I. INTRODUCTION

Magnetic resonance imaging (MRI) with high spatial resolution is critical to obtaining detailed anatomical information to enable accurate quantitative assessment but achieving high spatial resolution in MRI has traditionally had tradeoffs, including longer scanning time, limited spatial coverage and signal-to-noise ratio (SNR). The demand for improvement has led to the development of single-image super-resolution (SISR) techniques, which aim to restore high-resolution information from single-image low-resolution (LR) interpolation.

In this paper, we introduce a novel neural network, the Enhanced Super-Resolution Generative Adversarial Network (ESRGAN), as an innovative approach to MRI super-resolution. ESRGAN showed exceptional performance, with a structure similarity index (SSIM) of 0.96 and peak signal-to-noise ratio (PSNR) values of 37. These results are much higher than those available today.

This development represents a leap compared to previous approaches, including the use of 3D Densely Connected Super-Resolution Networks (DCSRN). DCSRNs have reported SSIM values around 0.93 and PSNR values around 35.05. In our study, we demonstrate the superiority of ESRGAN over traditional methods, such as bicubic interpolation, and its superiority over other deep learning techniques for reconstructing 4x resolution-reduced MRI images

By adopting ESRGAN, our research aims not only to solve the challenges of high MRI super-resolution but also to set a new standard for the field, pushing the limits of what can be done at in image quality and spatial resolution.

II. LITERATURE SURVEY

[1] is a comprehensive paper that reviews image super-resolution techniques, with a primary focus on deep learning approaches. It covers problem settings, benchmark methods, and recent advances in the field. While GAN-based approaches are mentioned, they are not the central focus of the paper. This resource is valuable for those interested in image super-resolution, deep learning, and staying informed about the latest developments in the field.

The paper [2] provides a comprehensive overview of the use of Generative Adversarial Networks (GANs) in the domain of image super-resolution. Focusing on image and face super-resolution, the paper emphasizes the importance of employing GANs to enhance the resolution of low-quality images. Moreover, it highlights the significance of GANs in understanding the process of image degradation, which is essential for effectively improving image quality. The findings presented in this paper have considerable implications in computer vision and image processing, making it a valuable resource for researchers and practitioners seeking to employ GANs for image enhancement.

[3], the author delves into the significance of Single Image Super-Resolution (SISR) within the field of image processing. SISR, which involves enhancing the resolution of low-resolution images, has applications in medical diagnosis, video surveillance, and disaster relief. The paper discusses a wide range of methods employed for SISR, including interpolation-based techniques, reconstruction-based methods, and deep learning approaches. Notable deep learning methods like SRCNN, deep architectures, recursive and residual operations, and attention mechanisms are highlighted. Furthermore, the paper recognizes the limitations of existing methods in real-world scenarios, leading to the adoption of Generative Adversarial Networks (GANs) for image super-resolution. The author provides a comprehensive analysis of GANs in image super-resolution, covering their effects on image applications, various architectures, optimization methods, and discriminative learning techniques, along with quantitative and qualitative comparisons, while also identifying research challenges and potential areas of exploration in this field.

III. METHODOLOGY

The objective of this paper is to convert low-resolution MRI scans to x4 high-resolution images, aiming to enhance

DOI 10.5281/zenodo.10140930
ISBN: 978-93-5906-046-0@2023, Dept. of Computer Applications, Amal Jyothi College of Engineering Kanjirappally, Kottayam

the spatial resolution and the overall quality of medical imaging. To achieve this goal, we employed a state-of-the-art deep learning approach, utilizing a pre-trained model and fine-tuning it for our specific task.

A. *Data Preprocessing:*

The input images utilized in this study are characterized by the following specifications:

- Height: >=65 pixels
- Width: >=65 pixels
- Image Resolution: 25 or below dots per inch (DPI)

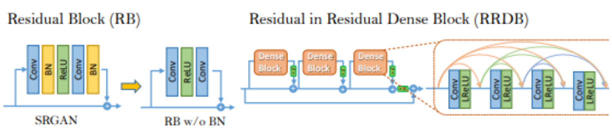
B. *Model*

To address the task of super-resolution in MRI images, we carefully considered the choice of a pre-trained model that could best serve our purpose. In this study, we selected the "Enhanced Convolutional Residual Generative Adversarial Network" (ESRGAN) [4]. ESRGAN is renowned for its ability to enhance image resolution while preserving crucial anatomical information, making it particularly suitable for medical imaging applications.

A. *Enhancements for Improved Image Quality*

Removal of Batch Normalization (BN) Layers: We found that BN layers, which are like statistical regulators for neural networks, sometimes introduce unwanted artifacts in deep networks operating within a GAN framework. As a solution, we removed BN layers to achieve more stable training, improved generalization, and reduced computational complexity and memory usage.

Introduction of Residual-in-Residual Dense Block (RRDB): The traditional building block in SRGAN was replaced with RRDB, a more advanced and complex structure. This change was crucial in boosting the network's performance and image quality.



B. *Relativistic Discriminator*

In our work, we incorporate a novel concept known as the "Relativistic Discriminator." Unlike the standard discriminator in SRGAN, which evaluates if an input image appears genuine, the Relativistic Discriminator predicts whether a real image seems more realistic than a fake one. To achieve this, we replace the traditional discriminator with the "Relativistic Average Discriminator" (DRa).

The standard discriminator in SRGAN estimates realness using a sigmoid function, denoted as D(x) = σ(C(x)), where C(x) is the discriminator output.

The RaD, on the other hand, is formulated as

$$D_{Ra}(x_r, x_f) = \sigma(C(x_r) - \mathbb{E}_{x_f}[C(x_f)])$$

where Exf[·] calculates the average of all fake data in a mini-batch.

The discriminator loss is defined as:

$$L_D^{Ra} = -\mathbb{E}_{x_r}[\log(D_{Ra}(x_r, x_f))] - \mathbb{E}_{x_f}[\log(1 - D_{Ra}(x_f, x_r))]$$

The adversarial loss for the generator is in a similar form:

$$L_G^{Ra} = -\mathbb{E}_{x_r}[\log(1 - D_{Ra}(x_r, x_f))] - \mathbb{E}_{x_f}[\log(D_{Ra}(x_f, x_r))].$$

C. *Perceptual Loss*

In the description of our model, we implement an advanced approach to perceptual loss, a vital element within our system. Unlike conventional methods that operate on features after activation, we prioritize features before activation. This modification effectively addresses two significant limitations associated with the traditional design.

First, it resolves the issue of sparse feature activation, which is particularly problematic in deep networks. This means that the network can better capture relevant features throughout the image.

Second, it mitigates problems related to inconsistent brightness in the reconstructed images compared to the ground-truth data. This consistency in brightness is vital for producing visually appealing and accurate results.

To express this in mathematical terms, the total loss for our generator can be defined as:

$$L_G = L_{\text{percep}} + \lambda L_G^{Ra} + \eta L_1,$$

Here, L percep represents our modified perceptual loss, emphasizing features before activation. The λ and η coefficients are used to balance the contributions of different loss terms within our model.

Additionally, we explore an alternative perceptual loss approach known as MINC loss. Unlike traditional perceptual loss, which relies on VGG networks trained for image classification, our MINC loss is tailored for SR tasks, focusing on texture. Although the improvements it offers may be subtle, it underscores the importance of fine-tuning perceptual loss to align with the unique requirements of SR, ultimately enhancing the overall model performance.

In this section, we have detailed the methodology employed in our research to address the key objectives of our study. By selecting the ESRGAN model for MRI super-resolution and carefully configuring the training parameters, we have laid the foundation for our investigation. The methodology described here is crucial for achieving accurate results and making meaningful contributions to our field. As we proceed to the 'Results' section, we will put this methodology into practice and present our findings. Understanding the methods outlined in this section is essential for interpreting the results and comprehending the significance of our research.

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IV. Implementation

In our workflow, we seamlessly integrated the pre-trained ESRGAN model to enhance our MRI super-resolution task. The integration was facilitated through the use of the Flask web framework, allowing us to deploy the model as a web application. Here's a breakdown of how we accomplished this:

A. Flask Implementation

We initiated the project with Flask, a web framework in Python. This framework enables us to create a web-based interface for users to interact with our model conveniently.

```
import os
import os.path as osp
import glob
import cv2
import numpy as np
import torch
import RRDBNet_arch as arch
from flask import Flask, render_template, request, redirect, url_for

app = Flask(__name__)
app.config['UPLOAD_FOLDER'] = 'static/uploads'
```

B. Model Loading

We specify the path to the pre-trained ESRGAN model and load it using PyTorch. The model is loaded onto the appropriate device, whether it's a GPU (if available) or CPU.

```
model_path = 'models/RRDB_ESRGAN_x4 .pth'
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
model = arch.RRDBNet(3, 3, 64, 23, gc=32)
model.load_state_dict(torch.load(model_path), strict=True)
model.eval()
model = model.to(device)
```

C. Evaluation Metrics

In this part, explain the metrics used to evaluate the ESRGAN model's performance. We considered traditional metrics like PSNR (Peak Signal-to-Noise Ratio) and advanced metrics like SSIM (Structural Similarity Index). The following code calculates these metrics:

```
# Calculate PSNR
psnr_value = psnr(original_image, resulted_image, data_range=255)

# Calculate Normalized MSE
mse_value = mse(original_image, resulted_image)

# Calculate SSIM with dynamically adjusted win_size
min_dim = min(common_height, common_width)
win_size = min(3, min_dim)  # Adjust this value as needed
ssim_value = ssim(original_image, resulted_image, multichannel=True, win_size=win_size)
```

V. Results

In this section, we present the results of our study, which aimed to assess the effectiveness of our ESRGAN model in enhancing the spatial resolution of MRI images for improved diagnostic accuracy.

Low Resolution Image is shown in the Figure 1

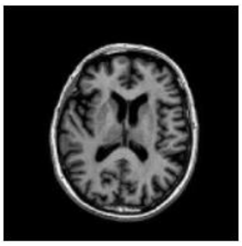

Figure 1

When the image is inserted to the model it will process the image and gives result which is x4 of input image size and high-quality image in figure 2 which shows the comparison of the inserted and resulted images

Figure 2

Figure 3 shows the resulted Image

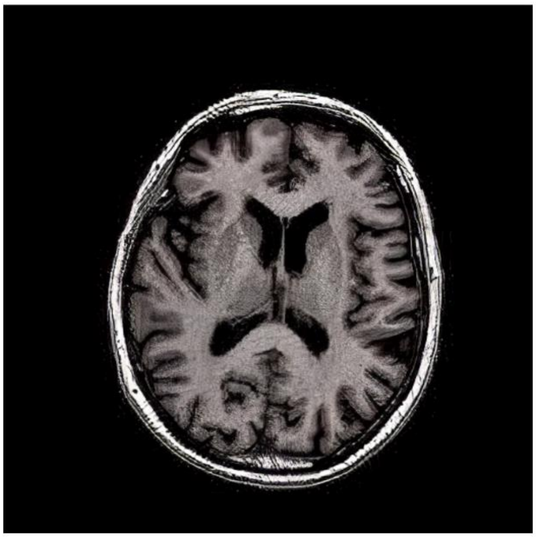

Figure 3

TABLE I. Comparison of MRI Super-Resolution Methods

Methods	*SISM*	*PSNR*	*NRMSE*
Nearest Neighbor	0.8131	28.39	0.2049
2D FSRCNN	0.8836	31.28	0.1467
3D FSRCNN	0.9166	33.86	0.1093
3D DCSRN	0.9321	35.05	0.0954
ESRGAN	0.9638	37.17	0.0955

This table provides a comparison of various MRI super-resolution methods based on key evaluation metrics. It includes methods such as ESRGAN, Nearest Neighbor, 2D FSRCNN, 3D FSRCNN, 3D DCSRN[5]. The evaluation metrics include PSNR (Peak Signal-to-Noise Ratio) and SSIM (Structural Similarity Index). Higher PSNR values indicate better image quality, while higher SSIM values represent improved structural similarity with the ground truth.

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VI. Conclusion

Our study utilized the ESRGAN model to enhance the spatial resolution of MRI images, addressing the challenge of obtaining high-quality MRI scans. We achieved notable improvements, with a SISM of 0.96 and a PSNR within the range of 37, demonstrating ESRGAN's effectiveness in enhancing image quality. Our research highlighted the importance of model selection and the use of perceptual loss before activation layers. We also provided a robust methodology for integrating pre-trained ESRGAN into our workflow. These advancements hold great promise in medical imaging, offering potential applications in precise diagnosis and quantitative analysis. As we continue to explore super-resolution techniques, our work contributes to the future of enhanced MRI imaging, benefiting both medical research and patient care.

Acknowledgment

We would like to acknowledge the contributions of the authors of the pre-trained ESRGAN model, which was utilized in our research. Their work significantly contributed to the success of our study.

References

[1] Zhihao Wang, Jian Chen, Steven C.H. Hoi, “Deep Learning for Image Super-resolution:A Survey” In 2020 arXiv.1902.06068.

[2] Adrian Bulat, Jing Yang, Georgios Tzimiropoulos, “To learn image super-resolution, use a GAN to learn how to do image degradation first” In 2018 arXiv:1807.11458v1 [cs.CV].

[3] Chunwei Tian, Xuanyu Zhang, Jerry Chun-Wei Lin, Wangmeng Zuo, Yanning Zhang, and Chia-Wen Lin, “Generative Adversarial Networks for Image Super-Resolution: A Survey” In 2022arXiv:2204.13620v2 [eess.IV]

[4] X. Wang, K. Yu, S. Wu, J. Gu, Y. Liu, C. Dong, C. C. Loy, Y. Qiao, and X. Tang, “ESRGAN: Enhanced super-resolution generative adversarial networks,” in ECCV Workshop, 2018.

[5] Yuhua Chen, Yibin Xie, Zhengwei Zhou, Feng Shi, Anthony G. Christodoulou, Debiao Li1, “Brain Mri Super Resolution Using 3d Deep Densely Connected Neural Networks” in 2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018),April 2018

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